

1. A 67-year-old woman comes to the physician because of a 1-month history of diffuse abdominal cramps and worsening constipation. Her last bowel movement was 4 days ago, and the stool was hard. She has a long-standing history of constipation. She typically has one bowel movement every other day, but this pattern varies from two bowel movements daily to one bowel movement weekly. She has stopped using a laxative because of diarrhea resulting in fecal incontinence. Colonoscopy 3 years ago showed no abnormalities. She has major depressive disorder with anxiety treated with sertraline and lorazepam. She does not appear to be in distress. Her temperature is 37.2°C (99°F), pulse is 88/min, respirations are 18/min, and blood pressure is 152/84 mm Hg. The abdomen is soft, nondistended, and nontender. Bowel sounds are decreased. Rectal examination shows no masses or stool; test of the stool for occult blood is negative. Which of the following is the most appropriate next step in management?
- ☐ A) X-rays of the abdomen
 - ☐ B) CT scan of the abdomen and pelvis
 - ☐ C) Discontinuation of lorazepam
 - ☐ D) Discontinuation of sertraline
 - ☐ E) Trial of acidophilus
 - ☐ F) Trial of fiber supplementation
 - ☐ G) Colonoscopy



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

2. A previously healthy 4-year-old girl is brought to the physician because of a sore throat and temperatures to 38.8°C (101.8°F) over the past 24 hours. She attends preschool, and several other students have similar symptoms. She appears ill. Examination shows erythema of the tonsils with exudate. The remainder of the examination shows no abnormalities. A rapid streptococcal antigen test is negative, but throat cultures grow *Streptococcus pyogenes* (group A). Which of the following characteristics of the rapid streptococcal antigen test best explains these results?

- ☐ A) Low likelihood ratio
- ☐ B) Low negative predictive value
- ☐ C) Low positive predictive value
- ☐ D) Low sensitivity
- ☐ E) Low specificity



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

3. A 42-year-old woman comes to the physician because of a 1-day history of severe pain, swelling, and warmth in her left knee. Her pain is moderately relieved with ibuprofen use. She notes that she awoke with the pain 1 day after she did an aerobic kickboxing workout. Her temperature is 37.2°C (99°F), pulse is 86/min, and blood pressure is 120/70 mm Hg. The upper aspect of the left knee is erythematous and warm to the touch with an effusion. Range of motion is limited by pain. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in management?
- ☐ A) Reassurance
 - ☐ B) Application of a knee brace and ice and elevation
 - ☐ C) MRI of the knees
 - ☐ D) Intravenous antibiotic therapy
 - ☐ E) Arthrocentesis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

4. A healthy 14-year-old girl is brought for a routine health maintenance examination. On questioning, she says that she has been sexually active with her boyfriend of 6 months. She says that they use condoms consistently and she takes an oral contraceptive. She has smoked one pack of cigarettes weekly for 1 year and drinks beer and wine occasionally at weekend parties. Physical examination shows no abnormalities. Which of the following is the most appropriate next step in management?
- ☐ A) Inform her about the long-term health consequences of smoking
 - ☐ B) Inquire about her attitudes toward smoking and health
 - ☐ C) Schedule a meeting to speak to her parents about her smoking
 - ☐ D) Prescribe a nicotine patch for smoking cessation
 - ☐ E) Prescribe sustained-release bupropion for smoking cessation



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

5. A healthy 52-year-old woman comes to the physician because she is concerned about her risk for developing thyroid cancer. Her father was just diagnosed with papillary carcinoma of the thyroid gland. She takes no medications. Examination shows no abnormalities. Which of the following is the most appropriate screening test for this patient?

- ☐ A) Palpation of the thyroid gland
- ☐ B) Measurement of serum thyroid antibodies
- ☐ C) ^{123}I scintigraphy of the thyroid gland
- ☐ D) Ultrasonography of the thyroid gland
- ☐ E) CT scan of the thyroid gland



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

6. A 19-year-old college student comes to the physician because of a 3-month history of cough that occurs only in midafternoon and evening and is not associated with any other symptoms. The cough seems to be somewhat worse in cold weather but does not wake her from sleep or prevent her from performing routine activities. She has no personal or family history of medical disease. She has never been hospitalized. Examination and an x-ray of the chest show no abnormalities. Which of the following is the most likely diagnosis?
- ☐ A) Asthma
 - ☐ B) Cocaine use
 - ☐ C) Granulomatosis with polyangiitis
 - ☐ D) Sarcoidosis
 - ☐ E) Tuberculosis
 - ☐ F) Vocal cord polyp



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

7. An 82-year-old man is brought to the physician by his daughter because of a 6-month history of memory problems and strange behavior. He lives alone. His daughter says that four times during this period, she found him wandering around his apartment naked in the middle of the day. She took his car and car keys 2 months ago to prevent him from driving because he repeatedly became lost while driving home from the grocery store. He has a history of head injury at the age of 14 years. He has hypercholesterolemia and hypertension. Current medications include simvastatin and diltiazem. His paternal grandmother had dementia, Alzheimer type. The patient has an eighth-grade education. Vital signs are within normal limits. Physical examination shows a slow but normal gait. On mental status examination, he describes his mood as "okay." He appears mildly irritated at having an appointment but is friendly. His Mini-Mental State Examination score is 22/30. Which of the following is the most appropriate next step in pharmacotherapy?

- ☐ A) Decrease the dosage of diltiazem
- ☐ B) Decrease the dosage of simvastatin
- ☐ C) Add clopidogrel to the current regimen
- ☐ D) Add donepezil to the current regimen
- ☐ E) Add selegiline to the current regimen
- ☐ F) Add sertraline to the current regimen



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

8. A 23-year-old woman comes to the physician because of moderate pain and swelling of her right ankle after twisting it yesterday. She has no history of serious illness and takes no medications. Vital signs are within normal limits. Examination shows moderate edema of the lateral aspect of the right ankle, an ecchymosis below the lateral malleolus, and tenderness to palpation of the anterior aspect of the lateral malleolus. The posterior aspects of the lateral and medial malleoli are nontender. Neurovascular function of the ankle is intact. The patient can bear weight as she walks across the room but has been using a set of crutches. In addition to recommending 2-day ice pack application to the injury, which of the following is the most appropriate next step in management?
- ☐ A) Stress x-rays of the ankle to assess ligament integrity
 - ☐ B) X-rays of the ankle to rule out fracture
 - ☐ C) Use of a soft protective brace and early range of motion exercises
 - ☐ D) Splinting the ankle and referral to an orthopaedist



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

9. A previously healthy 27-year-old woman, gravida 2, para 1, at 26 weeks' gestation comes to the physician for a routine prenatal visit. Her temperature is 36.7°C (98°F), pulse is 82/min, respirations are 20/min, and blood pressure is 120/70 mm Hg. Physical examination shows no abnormalities except for ecchymoses in various stages of healing over the forearms and lower abdomen. The fundal height is 27 cm. The cervix is long and closed, and the uterus is nontender and firm. Fetal heart tones are normal. Her hemoglobin concentration is 12 g/dL, platelet count is 280,000/mm³, and bleeding time is 6 minutes. Which of the following is the most likely diagnosis?

- ☐ A) Domestic abuse
- ☐ B) HELLP syndrome
- ☐ C) Immune thrombocytopenic purpura
- ☐ D) Preeclampsia
- ☐ E) von Willebrand disease



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

10. An 82-year-old woman comes to the physician because of difficulty sleeping at night for 6 months. She says she watches television or listens to talk radio at night because she cannot sleep. She has daytime fatigue and takes frequent naps. Her husband died 3 years ago, and she lost her pet dog of 11 years 8 months ago. She has had a 5.4-kg (12-lb) weight loss since her last examination 6 months ago. She is 173 cm (5 ft 8 in) tall and weighs 59 kg (130 lb); BMI is 20 kg/m². Her temperature is 37.5°C (99.5°F), pulse is 86/min, respirations are 10/min, and blood pressure is 150/84 mm Hg. She appears disheveled and has cracked, chapped lips. Physical examination shows no abnormalities. On mental status examination, her eye contact is poor. She answers questions with delayed, short responses and has difficulty finding the right words. She describes her activities during the past few days vaguely after being asked for specific details. When asked to draw the face of a clock and indicate 1:30 on the drawing, she requires 5 minutes to complete the clock face. She must be reminded that she was told to indicate 1:30, which she then proceeds to do correctly. She can recall two of three objects after 5 minutes. Which of the following is the most appropriate next step in management?
- ☐ A) Emphasize the importance of good sleep hygiene
 - ☐ B) Recommend grief counseling
 - ☐ C) Suggest she get a new household pet
 - ☐ D) Begin donepezil therapy
 - ☐ E) Begin sertraline therapy



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

11. A 16-year-old girl is brought to the physician by her parents because she has become increasingly withdrawn during the past 5 months. She quit the swim team and no longer enjoys spending time with friends. Her parents report that she was previously an honor roll student, but her grades have dropped to Cs. When questioned about her behavior, she states that she is tired and her stomach hurts. She sleeps 10 to 12 hours each night and takes a nap after school most days. She has had a 5.4-kg (12-lb) weight loss during this time that she attributes to decreased appetite. She reports that she is not concerned about her weight loss and is not intentionally dieting. She often has nausea after eating. She says she is not depressed or upset about anything, but she becomes tearful when discussing her physical symptoms saying, "Maybe I have mono or something. I am just so tired." She has not had fever, sore throat, cough, or diarrhea. Menses occur at regular 28-day intervals. Her last menstrual period was 1 week ago. She reports that she drank alcohol once at a party. She does not use illicit drugs. She has been sexually active with one male partner for the past 6 months, and they usually use condoms. She appears tired. She is 165 cm (5 ft 5 in) tall and weighs 55 kg (122 lb); BMI is 20 kg/m². Her temperature is 37.2°C (98.9°F), pulse is 92/min, respirations are 15/min, and blood pressure is 108/70 mm Hg. Examination shows no lymphadenopathy. Abdominal examination shows no abnormalities. Her hematocrit is 39%, and leukocyte count is 7000/mm³. Which of the following is the most appropriate next step in management?
- ☐ A) Assessment for suicide risk
 - ☐ B) Serologic testing for Epstein-Barr virus
 - ☐ C) Thyroid function tests
 - ☐ D) Urine pregnancy test
 - ☐ E) Endoscopy



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

12. A 77-year-old man comes to the physician because of a 4-week history of moderately severe aching and morning stiffness in his shoulders and hips. He has difficulty getting out of bed in the morning and reaching over his head because of the pain. During this period, he has had low-grade fever and generalized fatigue. Examination shows decreased range of motion of the shoulders and hips. There is no joint or muscle tenderness. His erythrocyte sedimentation rate is 70 mm/h. Which of the following is the most appropriate next step in management?

- ☐ A) Allopurinol therapy
- ☐ B) Amitriptyline therapy
- ☐ C) Colchicine therapy
- ☐ D) Cyclophosphamide therapy
- ☐ E) Ibuprofen therapy
- ☐ F) Methotrexate therapy
- ☐ G) Prednisone therapy
- ☐ H) Plasmapheresis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

13. A 56-year-old woman comes to the physician for a routine examination in October. She has type 2 diabetes mellitus well-controlled with metformin. Vital signs are within normal limits. Examination shows no abnormalities. The patient says that she does not want to get the influenza vaccine this year because she knows people who have "come down with the flu because of the shot." Which of the following is the most appropriate physician response?
- ☐ A) "As your physician, you should trust my recommendation. Please take the influenza vaccine."
 - ☐ B) "If you are not going to accept the vaccine, I'd like you sign a waiver of liability form."
 - ☐ C) "Influenza can cause pneumonia and death, and type 2 diabetes mellitus puts you at risk. How will your family feel if anything happens to you?"
 - ☐ D) "Patients who are incubating the influenza virus when they receive the vaccine mistakenly attribute their symptoms to the vaccine. You should get the vaccine."
 - ☐ E) "Why people still believe this myth I'll never know because scientific studies have proven that it is impossible for patients to get the flu from the vaccine."



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

14. An 82-year-old woman comes to the physician because of a 6-month history of frequent falls. Prior to each fall, she had a feeling of imbalance when walking or turning. She has never lost consciousness during these episodes. She has a 5-year history of stable coronary artery disease and a 10-year history of osteoarthritis of the knees. Current medications include 81-mg aspirin and lovastatin. She resides in an assisted living facility and has a bathroom with fixtures for elderly residents. Her pulse is 80/min, respirations are 14/min, and blood pressure is 140/85 mm Hg with no orthostatic changes. On examination, she has difficulty rising from her chair. Her gait while using her cane is slow with instability on turns. Muscle strength is 4/5 in the lower extremities. Which of the following is the most appropriate next step in management?

- ☐ A) Prescription for a powered wheelchair
- ☐ B) 24-Hour ambulatory ECG monitoring
- ☐ C) Duplex ultrasonography of the carotid arteries
- ☐ D) Physical therapy
- ☐ E) Meclizine therapy



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

15. A 32-year-old woman is brought to the emergency department by her husband because of the onset of shortness of breath, profuse sweating, and chest discomfort while at a concert 30 minutes ago. She had a similar episode 1 month ago while at her daughter's school play. She has felt anxious since that time and is concerned that she is going crazy. She has monitored her pulse during the past month, and it has ranged between 70–80/min. Examination shows a pulse of 110/min and blood pressure of 140/76 mm Hg. An opening systolic click is heard over the left fourth intercostal space. Which of the following is the most likely cause of her symptoms?

- ☐ A) Generalized anxiety disorder
- ☐ B) Hyperthyroidism
- ☐ C) Mitral valve prolapse
- ☐ D) Panic disorder
- ☐ E) Paroxysmal atrial tachycardia



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

16. A 42-year-old man comes to the physician for a routine follow-up examination. He feels well. He has a 2-year history of mild hypertension treated with sodium restriction and a reduced-calorie diet. He has had a 2-kg (4.4-lb) weight loss as a result of the diet. His father sustained a cerebral infarction at the age of 55 years, and the patient is concerned about his own risk for cerebral infarction. He does not smoke cigarettes and drinks two glasses of wine each night with dinner. He is an accountant and does not exercise regularly. He is 178 cm (5 ft 10 in) tall and weighs 123 kg (272 lb); BMI is 39 kg/m². His blood pressure is 160/98 mm Hg. Examination shows no other abnormalities. Serum lipid studies show a total cholesterol concentration of 228 mg/dL, an HDL-cholesterol concentration of 45 mg/dL, and an LDL-cholesterol concentration of 148 mg/dL. Which of the following is the strongest predisposing factor for cerebral infarction in this patient?

- ☐ A) Alcohol consumption
- ☐ B) Family history
- ☐ C) Hypercholesterolemia
- ☐ D) Hypertension
- ☐ E) Obesity
- ☐ F) Sedentary lifestyle



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

17. A 47-year-old woman comes to the physician because of a 6-month history of mild hand pain that occurs primarily over the palmar surfaces. The pain is associated with numbness and tingling, especially over her thumbs and middle and index fingers. Her symptoms are exacerbated by prolonged typing at work and sometimes awaken her from sleep. She has not had any difficulty writing by longhand or grasping small objects. She has type 2 diabetes mellitus well controlled with diet. She takes no medications. Vital signs are within normal limits. Tapping the palmar aspect of the wrist produces pain and tingling in both hands. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Use of wrist splints
- ☐ B) Oral nonsteroidal anti-inflammatory drug therapy
- ☐ C) Corticosteroid injections
- ☐ D) Endoscopic decompression of the median nerve
- ☐ E) Open decompression of the median nerve



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

18. A 52-year-old woman comes to the physician because of a 14-month history of generalized muscle pain and difficulty sleeping. The pain is most severe during periods of stress but is always present to some degree. She feels exhausted after awakening in the morning despite 8 to 9 hours of sleep nightly. Vital signs are within normal limits. Examination shows no abnormalities except for 12 areas of muscle point tenderness. In addition to recommending an aerobic exercise program, which of the following is the most appropriate treatment for this patient?

- ☐ A) Amantadine therapy
- ☐ B) Antidepressant therapy
- ☐ C) Benzodiazepine therapy
- ☐ D) Glucocorticoid therapy
- ☐ E) Opiate analgesic therapy
- ☐ F) Psychostimulant therapy



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



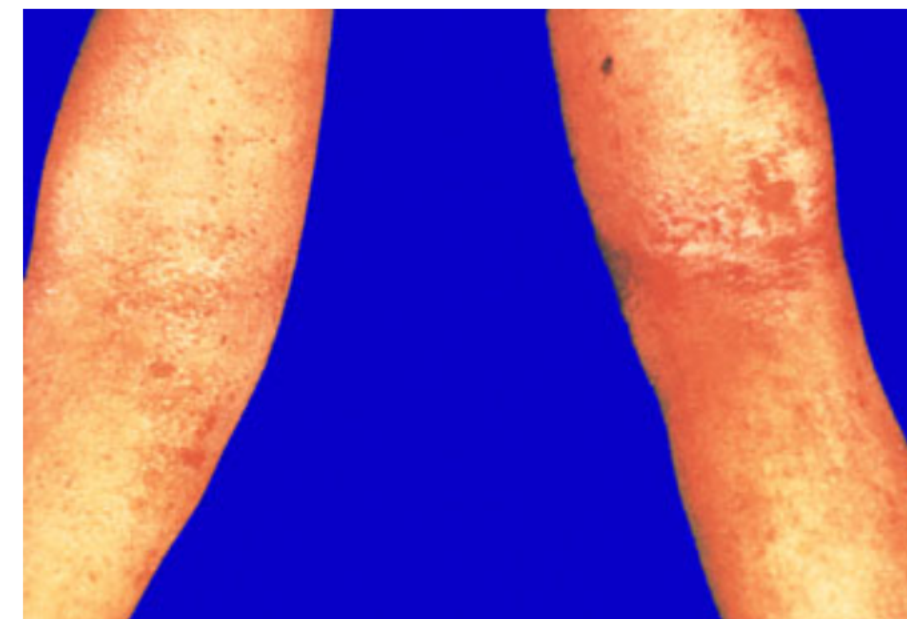
Help



Pause

19. An 18-year-old woman comes to the physician because of a pruritic facial rash for 2 days. The rash began around the lips and then spread to the cheeks. She is sexually active and uses an oral contraceptive. She has a history of asthma treated with a β -adrenergic agonist inhaler. There is a family history of extrinsic asthma. She has had a chronic rash on her extremities for 13 years; a photograph is shown. The uninvolved skin is chronically dry and scaling. Examination shows a vesicular facial rash and eczematous lichenified plaques on the cheeks, neck, and flexor surfaces of the extremities. Which of the following is the most likely diagnosis?

- ☐ A) Atopic dermatitis with eczema herpeticum
- ☐ B) Chronic contact dermatitis with superinfection
- ☐ C) Dermatitis herpetiformis
- ☐ D) Erythema multiforme with bullous formation
- ☐ E) Psoriasis with vesicular exacerbation



20. A previously healthy 2-year-old boy is brought to the physician by his mother because of hoarse cough and difficulty breathing that began yesterday evening and became progressively worse during the night. All appropriate immunizations are up-to-date. He appears tired but responds normally to stimuli. His temperature is 38°C (100.4°F), pulse is 164/min, respirations are 40/min, and blood pressure is 102/68 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 86%. Examination shows labored respirations and inspiratory stridor that can be heard without a stethoscope. Which of the following is the most likely diagnosis?

- ☐ A) Bacterial tracheitis
- ☐ B) Bronchiolitis
- ☐ C) Epiglottitis
- ☐ D) Laryngotracheobronchitis (croup)
- ☐ E) Retropharyngeal abscess



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

21. A 27-year-old man comes to the physician because of sadness, loss of appetite, and difficulty falling asleep since he and his wife separated 5 weeks ago. His energy level has decreased, and he no longer enjoys spending time with friends. He does not believe that his wife and he will be able to work through their conflicts but wishes they could. He has no history of serious illness. His sister has major depressive disorder well controlled with sertraline. The patient drinks one to two glasses of wine nightly. He does not use illicit drugs. He is 178 cm (5 ft 10 in) tall and weighs 104 kg (230 lb); BMI is 33 kg/m². Physical examination shows no abnormalities. On mental status examination, his mood is sad, and he is tearful. His thought process is logical. There is no evidence of suicidal ideation or hallucinations. Which of the following is the most appropriate treatment?

- ☐ A) Cognitive behavioral therapy
- ☐ B) Hypnotherapy
- ☐ C) Psychoanalysis
- ☐ D) Relaxation therapy
- ☐ E) Systematic desensitization



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

22. An asymptomatic 42-year-old man with polycystic kidney disease comes for a routine follow-up examination. He has hypertension treated with hydrochlorothiazide. His blood pressure has been 156/94 mm Hg on two previous visits. His temperature is 37°C (98.6°F), pulse is 80/min, and blood pressure is 160/96 mm Hg. The lungs are clear to auscultation. Cardiac examination shows a normal S₁ and S₂ with no murmurs, rubs, or gallops. His serum creatinine concentration is 1 mg/dL. Urinalysis shows 2+ protein and 5–10 RBC/hpf. Which of the following is the most appropriate next step in management?

- ☐ A) Echocardiography
- ☐ B) Intravenous pyelography
- ☐ C) Lisinopril therapy
- ☐ D) Trimethoprim-sulfamethoxazole therapy
- ☐ E) Cerebral arteriography



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

23. A 57-year-old woman comes to the physician for a routine health maintenance examination. She has no history of serious illness and takes no medications. She smoked one pack of cigarettes daily for 20 years but stopped 10 years ago. She drinks one glass of wine daily. She does not exercise. She is employed as an office assistant. She is 168 cm (5 ft 6 in) tall and weighs 65 kg (143 lb); BMI is 23 kg/m². Her pulse is 80/min, respirations are 14/min, and blood pressure is 140/90 mm Hg. The lungs are clear to auscultation. Cardiovascular examination shows no abnormalities. Serum lipid studies show:

Cholesterol, total	250 mg/dL
HDL-cholesterol	35 mg/dL
LDL-cholesterol	179 mg/dL
Triglycerides	180 mg/dL

Which of the following is the most appropriate next step in management?

- ☐ A) Exercise program
- ☐ B) Weight loss program
- ☐ C) Glucose tolerance test
- ☐ D) 24-Hour urine collection for measurement of microalbumin
- ☐ E) Serum lipid studies in 1 year
- ☐ F) Atenolol therapy



24. A 62-year-old man comes to the physician for a routine follow-up examination. He has type 2 diabetes mellitus and peripheral vascular disease. Current medications include metformin and glyburide. Dorsalis pedis and posterior tibial pulses are decreased bilaterally. Serum lipid studies show:

	One Year Ago	Today
Cholesterol, total (mg/dL)	190	180
HDL-cholesterol (mg/dL)	39	37
LDL-cholesterol (mg/dL)	139	135
Triglycerides (mg/dL)	180	140

Which of the following is the most appropriate recommendation?

- ☐ A) Repeat cholesterol testing in 3 months
- ☐ B) Dietary modification to decrease LDL-cholesterol concentration to less than 130 mg/dL
- ☐ C) Drinking wine with dinner to increase HDL-cholesterol concentration to greater than 40 mg/dL
- ☐ D) Lipid lowering therapy to decrease LDL-cholesterol concentration to less than 100 mg/dL
- ☐ E) No current treatment is indicated, and repeat cholesterol testing annually

25. A 77-year-old woman who is a resident of a skilled nursing care facility is brought to the physician because of a 1-week history of heartburn and difficulty swallowing. She has had no weight loss. She sustained a cerebral infarction 2 years ago. She has hypertension and hyperlipidemia. Current medications include chlorthalidone, potassium chloride, alendronate, aspirin, and lovastatin. Examination shows no abnormalities except for residual right-sided weakness. Upper endoscopy shows two shallow ulcers in the distal esophagus. Which of the following is the most likely diagnosis?

- ☐ A) Barrett esophagus
- ☐ B) Drug-induced esophagitis
- ☐ C) Esophageal cancer
- ☐ D) Herpes simplex esophagitis
- ☐ E) Reflux esophagitis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause



26. A previously healthy 6-year-old boy comes to the physician because of a 1-week history of an itchy, nonpainful rash on his arm. His mother says that it began as a small red spot but has tripled in size since then. He has had no other symptoms. A photograph of the lesion is shown. Which of the following is the most appropriate topical pharmacotherapy?

- ☐ A) Clindamycin
- ☐ B) Clotrimazole
- ☐ C) Hydrocortisone
- ☐ D) Permethrin
- ☐ E) Selenium sulfide





27. A 16-year-old boy is brought to the physician because of a 1-year history of severe acne that has not responded to treatment with over-the-counter topical medications. He has no history of serious illness and takes no other medications. He works in a garage on weekends and participates on the track team at school. He occasionally drinks coffee in the morning and sodas after work. He is 178 cm (5 ft 10 in) tall and weighs 64 kg (142 lb); BMI is 20 kg/m². A photograph of the forehead is shown. Genital and pubic hair development are Tanner stage 5. In addition to oral antibiotic and topical retinoid therapy, which of the following is the most appropriate recommendation for this patient?

- ☐ A) Avoidance of caffeine
- ☐ B) Avoidance of sun exposure
- ☐ C) Cleansing with topical alcohol twice daily
- ☐ D) Low-fat diet
- ☐ E) Use of moisturizers three times daily



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

28. A 32-year-old man comes to the physician because of painful blisters on his penis for 2 days. During the past 2 years, he has had similar blisters that occur approximately four times annually and last 10 days. Six months ago, a viral culture of fluid from a blister was negative. He currently takes no medications. He does not smoke or drink alcohol. He has had six lifetime female sexual partners and does not use condoms. His current sexual partner is asymptomatic. His temperature is 38.1°C (100.6°F), pulse is 80/min, respirations are 12/min, and blood pressure is 130/80 mm Hg. There is bilateral inguinal lymphadenopathy. Examination of the penis shows multiple 3-mm vesicular lesions on the shaft. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Acyclovir
- ☐ B) Ceftriaxone
- ☐ C) Doxycycline
- ☐ D) Ibuprofen
- ☐ E) Penicillin
- ☐ F) Prednisone



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



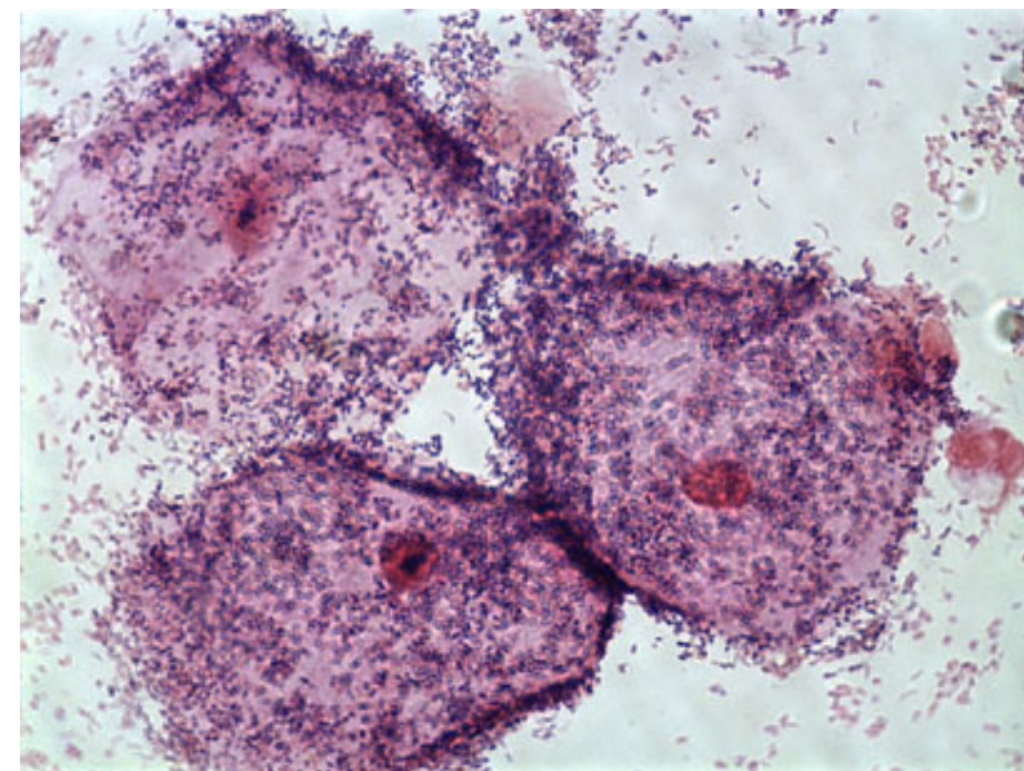
Help



Pause

29. A previously healthy 22-year-old woman comes to the physician because of a 3-day history of malodorous vaginal discharge. She has not had fever, chills, or pain with urination. Menses occur at regular 28-day intervals. Her last menstrual period was 2 weeks ago. She uses an oral contraceptive and has not missed a dose. Her temperature is 37°C (98.6°F), pulse is 78/min, respirations are 16/min, and blood pressure is 130/76 mm Hg. On abdominal examination, bowel sounds are normal. Pelvic examination shows gray, thin, watery vaginal discharge but no other abnormalities. Urinalysis and a pregnancy test are negative. A photomicrograph of a Gram stain of the discharge is shown. Results of cervical testing for *Chlamydia trachomatis* and *Neisseria gonorrhoeae* are pending. Which of the following is the most appropriate next step in management?

- ☐ A) Ceftriaxone therapy
- ☐ B) Doxycycline therapy
- ☐ C) Fluconazole therapy
- ☐ D) Metronidazole therapy
- ☐ E) No treatment is indicated until cervical testing results are received



30. An 82-year-old man comes to the physician for a follow-up examination. He has a 2-year history of a painful tingling sensation in his feet that did not respond to trials of gabapentin and carbamazepine. He has a 20-year history of type 2 diabetes mellitus treated with glyburide. Examination today shows normal muscle strength. Sensation to light touch is decreased to the midcalves bilaterally. Which of the following is the most appropriate alternative pharmacotherapy for this patient?

- ☐ A) Fluoxetine
- ☐ B) Morphine
- ☐ C) Nortriptyline
- ☐ D) Phenelzine
- ☐ E) Trazodone



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

31. A 42-year-old woman comes to the physician because of recent high blood pressure readings. During the past month, her blood pressure was 150/94 mm Hg on two occasions at health fairs. She has a history of migraines that have occurred twice weekly during the past 6 months. She has been taking ibuprofen four to six times weekly for her migraines. She has no other history of serious illness and takes no other medications. Her pulse is 88/min, and blood pressure is 152/92 mm Hg. The remainder of the examination shows no abnormalities. In addition to discontinuing ibuprofen therapy, administration of which of the following is the most appropriate next step in pharmacotherapy?

- ☐ A) ACE inhibitor
- ☐ B) α -Adrenergic blocking agent
- ☐ C) β -Adrenergic blocking agent
- ☐ D) Angiotensin-receptor blocking agent
- ☐ E) Daily aspirin



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

32. A previously healthy 30-year-old woman comes to the physician because of fever, nonproductive cough, and left-sided chest pain for 5 days; her symptoms started during a hiking trip with several friends in late August. One of these friends has similar symptoms. Mild diffuse crackles are heard. Which of the following is the most likely causal organism?

- ☐ A) *Borrelia burgdorferi*
- ☐ B) Coxsackievirus
- ☐ C) Influenza virus
- ☐ D) *Mycoplasma pneumoniae*
- ☐ E) *Streptococcus pneumoniae*



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

33. A previously healthy 32-year-old man comes to the physician because of a 4-month history of constant pain on the outside of his right elbow at rest. The pain has not responded to use of nonsteroidal anti-inflammatory drugs or a wrist splint. He is right-handed. He recently began building an addition to his house, and has been lifting heavy supplies. He is 175 cm(5 ft 9 in) tall and weighs 75 kg(165 lb); BMI is 24 kg/m². Examination shows severe tenderness of the bony right distal humerus 1 cm proximal to the radial head. Extension of the right wrist against resistance produces pain at the area of maximal tenderness. AP and lateral x-rays of the right elbow show no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Use of a forearm strap
- ☐ B) Oral vitamin B₆ supplementation
- ☐ C) Oral corticosteroid therapy
- ☐ D) Intravenous corticosteroid therapy
- ☐ E) Surgical decompression



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

34. A 37-year-old man comes to the physician because of intermittent chest pain over the past 2 days. Four days ago, he had a sore throat, headache, and malaise. Two days ago, he developed fever and a sharp stabbing pain in the right lateral chest. The pain resolved spontaneously after 1 hour but has recurred six times since then. There is no pain with exertion. He appears well. His temperature is 37°C (98.6°F), pulse is 80/min, and blood pressure is 120/70 mm Hg. The lungs are clear to auscultation. Cardiac examination shows a normal S₁ and S₂ with no murmurs, rubs, or gallops. Laboratory studies show:

Hematocrit	45%
Leukocyte count	3700/mm ³
Platelet count	250,000/mm ³
Serum creatine kinase	500 U/L

An x-ray of the chest and an ECG show no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Empyema
- ☐ B) Pericarditis
- ☐ C) Pleurodynia
- ☐ D) Pneumothorax
- ☐ E) Pulmonary embolism
- ☐ F) Systemic lupus erythematosus



35. A 37-year-old woman comes to the physician because of a 3-month history of generalized fatigue. She has noticed a decreased energy level, which is most pronounced at the end of the day. She has a 3-year history of abdominal bloating and intermittent diarrhea, especially after eating pasta. Her appetite has not changed, and she has had no weight loss. Menses occur at regular 28-day intervals with light flow for 2 days. She is 168 cm (5 ft 6 in) tall and weighs 70 kg (155 lb); BMI is 25 kg/m². Her pulse is 80/min, and blood pressure is 110/80 mm Hg. Examination shows moderately pale oral mucosa. The abdomen is distended and tympanitic; there is no tenderness or organomegaly. Rectal examination shows no abnormalities. Test of the stool for occult blood is negative. Laboratory studies show:

Hematocrit	30%
Mean corpuscular volume	78 μm ³
Leukocyte count	6400/mm ³
Platelet count	450,000/mm ³

Which of the following is most likely to confirm the underlying cause of her symptoms?

- ☐ A) Measurement of serum folic acid concentration
- ☐ B) Schilling test
- ☐ C) Serum *Helicobacter pylori* antibody assay
- ☐ D) CT scan of the pancreas
- ☐ E) Colonoscopy
- ☐ F) Assay for anti-transglutaminase antibodies

36. An 11-year-old boy is brought to the camp infirmary by his cabinmates 1 hour after collapsing. The boys had been playing a game that involved standing, breathing deeply and rapidly for 1 minute, and then expiring against a closed mouth. The patient collapsed and struck his head on the edge of his bed. He did not open his eyes or move for several minutes but is now alert and fully oriented. There is a 1 x 1-cm ecchymosis on the occipital scalp. His pupils react to light. Reflexes are symmetric; gait and muscle strength are normal. Which of the following is the most likely cause of the loss of consciousness?

- ☐ A) Akinetic seizure
- ☐ B) Cardiac arrhythmia
- ☐ C) Cerebral hypoperfusion
- ☐ D) Epidural hematoma
- ☐ E) Hypercarbia



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

37. A 27-year-old man comes to the physician because of a 1-week history of yellow eyes and dark urine. He has a history of intravenous drug use. Laboratory studies 6 months ago showed a positive hepatitis A antibody assay, a negative hepatitis B surface antigen assay, and a positive hepatitis B surface antibody assay. Examination today shows jaundice and hepatomegaly. Laboratory studies are consistent with hepatitis. Which of the following is the most likely cause of his hepatitis?

- ☐ A) Delta virus infection
- ☐ B) Hepatitis C infection
- ☐ C) Reactivation of hepatitis A
- ☐ D) Reinfection with hepatitis A
- ☐ E) Reinfection with hepatitis B



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



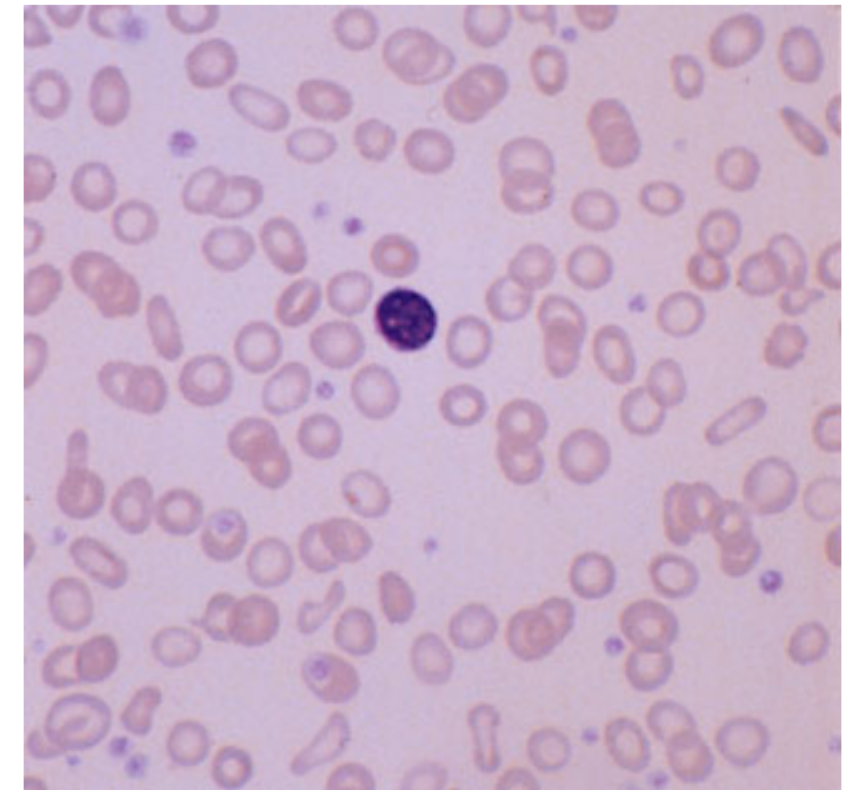
Pause

38. A 32-year-old woman is referred to the physician because of anemia noted on routine testing prior to blood donation. She felt well until recently when she began to experience decreased exercise tolerance. She has regular menstrual periods with no abnormal bleeding. Examination shows no abnormalities. Test of the stool for occult blood is negative. Laboratory studies show:

Hematocrit	27%
Mean corpuscular volume	73 μm^3
Leukocyte count	5200/ mm^3 with a normal differential

A blood smear is shown. Which of the following is the most appropriate initial step in management?

- ☐ A) Administration of folic acid
- ☐ B) Parenteral iron therapy
- ☐ C) Measurement of serum iron and ferritin concentrations
- ☐ D) Hemoglobin electrophoresis
- ☐ E) Bone marrow aspiration



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

39. A previously healthy 37-year-old woman who works as a prison guard comes to the physician because of a 2-month history of low-grade fever, fatigue, and a cough productive of scanty white sputum that is occasionally blood-streaked. She has had no shortness of breath. She has smoked one pack of cigarettes daily for 20 years and drinks four alcoholic beverages daily. Her temperature is 37.6°C (99.7°F), pulse is 88/min, respirations are 16/min, and blood pressure is 95/68 mm Hg. Examination shows rhonchi that clear with coughing; no crackles are heard. Examination shows no other abnormalities. Laboratory studies show:

Hemoglobin	9.5 g/dL
Serum	
Bilirubin	1.2 mg/dL
Alkaline phosphatase	80 U/L
AST	75 U/L
ALT	60 U/L

An x-ray of the chest shows a normal cardiac silhouette and a right upper lung field infiltrate; there are no pleural effusions. Which of the following is the most likely cause of the cough?

- ☐ A) Bronchitis
- ☐ B) Environmental irritants
- ☐ C) Lung cancer
- ☐ D) Pneumonia
- ☐ E) Sarcoidosis
- ☐ F) Tuberculosis

40. An otherwise healthy 9-year-old girl is brought to the physician by her mother because she noticed a lump in her daughter's left breast 10 days ago. The mother has a history of breast cancer. The patient is at the 70th percentile for height and 40th percentile for weight. Breast development is Tanner stage 2, and pubic hair development is Tanner stage 1. There is no axillary lymphadenopathy. Examination shows a 2 × 2.5-cm slightly raised mass just beneath the left areola. The mass is tender and slightly granular; there is no erythema. The right nipple is slightly tender but otherwise normal. Which of the following is the most appropriate next step in management?

- ☐ A) Follow-up examination in 6 months
- ☐ B) Mammography
- ☐ C) Ultrasound-directed biopsy of the mass
- ☐ D) Excision of the mass
- ☐ E) Antibiotic therapy



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

41. Over the past 4 months, a 66-year-old woman has had progressive shortness of breath on exertion. For 2 weeks, she has had two-pillow orthopnea and paroxysmal nocturnal dyspnea not associated with neck, chest, or arm pain. She has a history of untreated borderline hypertension. Her pulse is 96/min, respirations are 18/min, and blood pressure is 150/94 mm Hg. Examination shows 3-cm jugular venous distention. Bilateral inspiratory crackles are heard most prominently at both lung bases. There is a laterally displaced apical impulse and an S₃ gallop without murmur. There is mild pitting edema bilaterally. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Measurement of creatinine clearance
- ☐ B) Serial measurement of serum cardiac enzyme activities
- ☐ C) Measurement of left ventricular ejection fraction
- ☐ D) Thallium stress test
- ☐ E) Coronary artery angiography



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

42. A 16-year-old girl comes to the physician because of a 2-day history of urinary frequency and urgency and pain with urination. Over the past 2 months, she has had two urinary tract infections treated with trimethoprim-sulfamethoxazole. Menses occur at regular intervals, and she frequently uses tampons. She became sexually active with one partner 3 months ago. She uses depot medroxyprogesterone for contraception. She and her partner use condoms consistently, and they both have tested negative for sexually transmitted diseases. Examination shows bilateral pelvic tenderness to palpation and no costovertebral angle tenderness. After treatment for the current infection, which of the following is the most appropriate recommendation to prevent future urinary tract infections in this patient?
- ☐ A) Delay postcoital urination
 - ☐ B) Discontinue tampon use
 - ☐ C) Postcoital single-dose nitrofurantoin therapy
 - ☐ D) Switch from depot medroxyprogesterone to a diaphragm
 - ☐ E) Switch from depot medroxyprogesterone to an oral contraceptive



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

43. A 42-year-old woman comes to the physician for a follow-up examination. She has had persistent right heel pain since taking a 5-mile walk 3 weeks ago. She has been treated with naproxen, a heel cushion, and exercises. The pain has persisted, and she had the onset of foot swelling 4 days ago. She has well-controlled type 1 diabetes mellitus and hypertension. Her only other medications are benazepril and insulin. Examination shows 1+ edema bilaterally to the midcalf. Which of the following is the most likely cause of the edema?

- ☐ A) Decreased cardiac output
- ☐ B) Idiopathic anaphylactoid reaction
- ☐ C) Impaired renal excretion of sodium
- ☐ D) Localized soft-tissue swelling
- ☐ E) Naproxen-induced hepatic toxicity



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

The following vignette applies to the next **2** items. The items in the set must be answered in sequential order. Once you click **Proceed to Next Item**, you will not be able to add or change an answer.

A 55-year-old man comes to the physician for a routine health maintenance examination. He is asymptomatic. He has no history of serious illness and takes no medications. His father has hypertension, and his mother has type 2 diabetes mellitus. The patient has smoked one and one-half packs of cigarettes daily for 35 years. He drinks four beers daily. He considers his alcohol consumption normal and states it has never caused him personal or work problems. He says he has never drunk alcohol in the morning to improve hangover symptoms. He eats a high-fat diet. He has a sedentary lifestyle. He is 180 cm (5 ft 11 in) tall and weighs 109 kg (240 lb); BMI is 34 kg/m². His temperature is 36.6°C (97.9°F), pulse is 82/min, respirations are 16/min, and blood pressure is 136/88 mm Hg. The remainder of the examination shows no abnormalities.

44. **Item 1 of 2**
- Which of the following is the most appropriate screening test for this patient?
- ☐ A) Exercise stress test
 - ☐ B) 5-Hour glucose tolerance test
 - ☐ C) Measurement of serum prostate-specific antigen concentration
 - ☐ D) Serum lipid studies
 - ☐ E) Ultrasonography to screen for abdominal aortic aneurysm

Proceed to Next Item

You must select "Proceed to Next Item" to view Item 2 of 2

The following vignette applies to the next **2** items. The items in the set must be answered in sequential order. Once you click **Proceed to Next Item**, you will not be able to add or change an answer.

A 55-year-old man comes to the physician for a routine health maintenance examination. He is asymptomatic. He has no history of serious illness and takes no medications. His father has hypertension, and his mother has type 2 diabetes mellitus. The patient has smoked one and one-half packs of cigarettes daily for 35 years. He drinks four beers daily. He considers his alcohol consumption normal and states it has never caused him personal or work problems. He says he has never drunk alcohol in the morning to improve hangover symptoms. He eats a high-fat diet. He has a sedentary lifestyle. He is 180 cm (5 ft 11 in) tall and weighs 109 kg (240 lb); BMI is 34 kg/m². His temperature is 36.6°C (97.9°F), pulse is 82/min, respirations are 16/min, and blood pressure is 136/88 mm Hg. The remainder of the examination shows no abnormalities.

44. Item 1 of 2

Which of the following is the most appropriate screening test for this patient?

- ☐ A) Exercise stress test
- ☐ B) 5-Hour glucose tolerance test
- ☐ C) Measurement of serum prostate-specific antigen concentration
- ☐ D) Serum lipid studies
- ☐ E) Ultrasonography to screen for abdominal aortic aneurysm

45. Item 2 of 2

Results of serum lipid studies are within the reference ranges. During a follow-up examination 2 weeks later, the patient's blood pressure is 138/88 mm Hg. Physical findings are unchanged. Which of the following historical findings represents the strongest mortality risk factor for this patient during the next 10 years?

- ☐ A) Blood pressure
- ☐ B) High-fat diet
- ☐ C) Sedentary lifestyle
- ☐ D) Smoking
- ☐ E) Weight

End of Set



Previous



Next

<https://t.me/USMLENBME2CK>



Lab Values



Calculator



Review



Help



Pause

46. A 62-year-old woman comes to the physician because of a 9-month history of pain and swelling of her finger joints. Her pain increases with activity. She has had progressive stiffness in her fingers during the past 2 years. She has mild hypertension controlled with diet. She takes no medications. Her blood pressure is 138/90 mm Hg. Examination shows an asymmetric distribution of enlarged bony proximal and distal interphalangeal joints; some of the joints are tender to palpation, and passive range of motion is limited by pain. Grip strength is normal. The remainder of the examination shows no abnormalities. The most appropriate initial step in management is administration of which of the following?

- ☐ A) Acetaminophen
- ☐ B) Allopurinol
- ☐ C) Gold
- ☐ D) Methotrexate
- ☐ E) Prednisone



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

47. A 9-month-old girl is brought to the physician for a well-child examination. Two weeks ago, she had viral gastroenteritis; her symptoms lasted for 10 days and included low-grade fever, vomiting, and diarrhea. Her mother recently resumed feeding her daughter the cow milk–based formula she had been drinking before the illness. The infant's stools have continued to be watery and she has had "a lot of gas." She is otherwise healthy, and her immunizations are up-to-date. Vital signs are within normal limits. Physical examination shows no abnormalities except for mild abdominal distention with increased bowel sounds. There are no palpable masses or organomegaly. Which of the following is the most likely explanation for this patient's current gastrointestinal symptoms?

- ☐ A) Bacterial enteritis
- ☐ B) Celiac disease
- ☐ C) Cow milk allergy
- ☐ D) Lactose intolerance
- ☐ E) Persistent viral gastroenteritis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

48. A 47-year-old woman comes to the physician to discuss weight loss. She has a 5-year history of hypertension and was recently diagnosed with type 2 diabetes mellitus. She has been overweight her entire life and gained approximately 9 kg (20 lb) after the birth of each of her three children. She has tried over-the-counter medications for weight loss and lost 18 kg (40 lb) on commercial diets but has been unable to keep the weight off. She is not following any particular diet at this time but she walks for 30 minutes three times weekly. Current medications are hydrochlorothiazide and metformin. Her father died of a myocardial infarction at the age of 52 years. She does not smoke cigarettes. She is 165 cm (5 ft 5 in) tall and weighs 104 kg (230 lb); BMI is 38 kg/m². Her temperature is 37°C (98.6°F), pulse is 84/min, respirations are 12/min, and blood pressure is 140/80 mm Hg. Examination shows no other abnormalities. Serum studies show:

Glucose	140 mg/dL
Cholesterol, total	225 mg/dL
HDL-cholesterol	50 mg/dL
LDL-cholesterol	120 mg/dL
Triglycerides	280 mg/dL

A complete blood count and serum electrolyte concentrations are within the reference range. Liver and thyroid function tests show no abnormalities. Which of the following is most likely to be effective in achieving weight loss in this patient?

- ☐ A) Exercise program
- ☐ B) Gastric bypass
- ☐ C) Low-carbohydrate diet
- ☐ D) Low-fat diet
- ☐ E) Orlistat therapy

49. An asymptomatic 67-year-old woman comes to the physician for a follow-up visit. She has a 10-year history of type 2 diabetes mellitus and a 3-year history of hypertension. Her medications are glyburide, metoprolol, amlodipine, and fluoxetine. Her blood pressure is 136/86 mm Hg. The lungs are clear to auscultation. S₁ and S₂ are normal. The point of maximal impulse is at the fifth intercostal space at the midclavicular line. Examination of the lower extremities shows 2+ edema from the mid calf to the foot bilaterally. Sensation to pinprick and vibration is decreased over the feet. Laboratory studies show:

Serum		
Urea nitrogen		8 mg/dL
Creatinine		0.8 mg/dL
Urine		
Glucose		1+
Protein		trace

Which of the following is the most likely cause of this patient's edema?

- ☐ A) Adverse effect of medication
- ☐ B) Congestive heart failure
- ☐ C) Deep venous thrombosis
- ☐ D) Lymphatic obstruction
- ☐ E) Nephrotic syndrome

50. The mother of a 16-year-old boy telephones the physician to request drug screening during his annual examination. She suspects that he has been using drugs because his grades recently dropped, and she has heard rumors that some of his friends are using drugs. She says that she has not discussed her request with her son and asks the physician to perform the drug screening without her son's knowledge. Which of the following is the most appropriate course of action?
- ☐ A) Agree to testing to abide by the mother's wishes
 - ☐ B) Agree to testing after she faxes written documentation of her request to the office
 - ☐ C) Agree to testing but inform her that telling her the results would violate the physician-patient relationship
 - ☐ D) Do not agree to testing because her son is an emancipated minor and she cannot legally make such a request
 - ☐ E) Do not agree to testing unless her son agrees to the drug testing



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause